

Blown trailing edge: how much lift does the jet buy?

Two-dimensional aerofoil section, jet blown from the trailing edge at $\tau = 30^\circ$ below the chord line, $\alpha = 0^\circ$, chord Reynolds number 1.0×10^6 , chord 1.0 m, free stream 10.0 m s^{-1} .

The answer in one line

Lift does not grow in proportion to blowing. It grows as the *square root* of the jet momentum coefficient C_μ , so the first increment of blowing is worth far more than the last: the first 0.05 of C_μ buys 8.47 of lift coefficient per unit C_μ , the last 0.20 only 1.95 — a factor of 4.3. Sizing a blowing system on a straight line through a low-blowing data point overpredicts the lift at $C_\mu = 0.40$ by 169%.

Table 1 – Published thin-aerofoil theory for a jet-flapped section

Jet momentum coefficient C_μ [-]	Total lift coefficient $(C_L)_\infty$ [-]	of which: jet reaction $C_\mu \sin(\tau + \alpha)$ [-]	pressure lift [-]	Marginal lift per unit C_μ dC_L/dC_μ [-]	Uncertainty on the value [-]
0.00	0	0	0	—	$< 10^{-9}$
0.05	4.23403×10^{-1}	2.50000×10^{-2}	3.98403×10^{-1}	4.331	$< 10^{-9}$
0.10	6.04776×10^{-1}	5.00000×10^{-2}	5.54776×10^{-1}	3.131	$< 10^{-9}$
0.20	8.68742×10^{-1}	1.00000×10^{-1}	7.68742×10^{-1}	2.294	$< 10^{-9}$
0.40	1.25948×10^0	2.00000×10^{-1}	1.05948×10^0	1.716	$< 10^{-9}$

The jet-reaction column is the direct vertical component of the jet momentum, reported separately and never folded silently into the lift column: here it is a *part* of the total, and the pressure-lift column is what remains. The marginal column is the slope of the total with respect to C_μ ; it is unbounded as $C_\mu \rightarrow 0$, which is the square-root law stated as a number. Uncertainty is the arithmetic error of evaluating the published expression; the accuracy of the published fit is not stated in the source.

Table 2 – Solved flow field: four blown conditions and a slot-closed reference

Jet momentum coefficient C_μ [-]	Lift from the aerofoil surface $C_{L,surface}$ [-]	Jet reaction (added, not measured) [-]	Lift, surface + jet reaction [-]	Movement in the last 4,000 iterations [-]	Turbulence equation imbalance [-]	Difference from Table 1 [%]
0.00 [†]	5.50595×10^{-5}	0	5.50595×10^{-5}	7×10^{-7}	3.46×10^{-6}	—
0.05	4.05733×10^{-1}	2.50000×10^{-2}	4.30733×10^{-1}	5×10^{-7}	5.94×10^{-6}	+1.73
0.10	5.48846×10^{-1}	5.00000×10^{-2}	5.98846×10^{-1}	7×10^{-7}	2.55×10^{-5}	-0.98
0.20	7.43999×10^{-1}	1.00000×10^{-1}	8.43999×10^{-1}	4×10^{-5}	4.57×10^{-5}	-2.85
0.40	1.00999×10^0	2.00000×10^{-1}	1.20999×10^0	8×10^{-6}	1.47×10^{-4}	-3.93

The surface column integrates pressure and shear over the aerofoil only; it does *not* contain the jet reaction, which is added explicitly in the fourth column to give a quantity comparable with Table 1. “Movement” is how much the lift still shifted over the final 4,000 iterations: half the range it covered there, rounded up to one significant figure. “Turbulence equation imbalance” is how far the turbulent kinetic energy equation is from being satisfied at the last iteration; the target set before running was 1×10^{-6} . [†]The $C_\mu = 0$ row is a reference case with the slot *closed*; the other four have it open and differ only in jet strength.

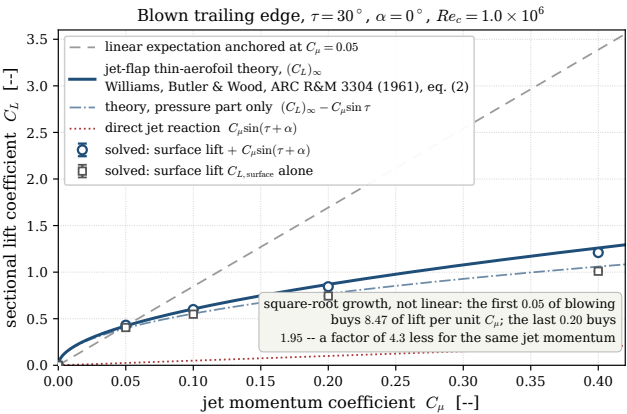


Figure 1. Lift against blowing. The solid curve is published thin-aerofoil theory for a jet-flapped section (Williams, Butler & Wood, ARC R&M 3304, 1961, eq. 2). The dashed grey line is what a linear expectation anchored at $C_\mu = 0.05$ would have predicted; the gap between the two is the message of this sheet. Uncertainty bars on the solved points are the movement column of Table 2 and are smaller than the markers.

What has been checked, what has not, and how the team reads it

Checked. The theory curve reproduces the published expression exactly. The lift derivatives were read off the printed page of Williams, Butler & Wood, *The Aerodynamics of Jet Flaps*, Aeronautical Research Council Reports and Memoranda No. 3304 (H.M.S.O., 1963), page 5, equation (2), and the jet-reaction convention $C_\mu \sin(\tau + \alpha)$ off page 3. Our evaluation reproduces the values fixed before running to better than 5×10^{-10} in lift coefficient at every point in Table 1, and the marginal-lift column matches an independent finite-difference evaluation of the same expression to eight figures.

Checked. The reference section, slot closed, returns a surface lift coefficient of 5.51×10^{-5} at zero incidence, as a symmetric section must. The same solver and reader gave 1.01 at the strongest blowing, so the near-zero is physics, not a reader that could only give zero.

Not checked. None of the solved points in Table 2 is offered as verified against anything. The solutions are still tightening: no condition has brought every equation below the 1×10^{-6} target set before running, and no mesh-refinement study has been completed for these five conditions. The last column of Table 2 is an observation, not an agreement claim.

Lead Researcher. The physics worth taking away is the square-root

Please read before using these numbers.

- The theory curve is trustworthy arithmetic against a published formula. The solved points are not yet finished calculations and none of them is described here as verified.
- The uncertainty column in Table 2 says how much each number was still moving at the end of the run. It is *not* a total error bar: the contribution from mesh resolution has not been established for these five conditions, and it is larger than the number shown.
- The published formula is an interpolation fit to computed values, valid for blowing from the trailing edge with no separate flap, and is used whole over the range it was fitted for. It is inviscid thin-aerofoil theory: no stall, no separation, no viscous loss, so it overstates lift wherever the real jet sheet or boundary layer breaks down.
- Everything here is two-dimensional. No finite-span correction has been applied, and a real wing of finite aspect ratio will achieve less.

What this cost

The five solved conditions used 117.5 core-minutes of compute in total (23.8, 24.4, 21.8, 22.3, 25.2 core-minutes respectively). The figure quoted before any of them started was 56.8 core-minutes, so the work came in 2.07 times the upfront estimate. The sixth, finer-mesh companion condition was quoted at 90.5 core-minutes upfront and used 98.7, 1.09 times its estimate. The theory curve costs no solver time. At the machine's hourly rate of \$0.0513 per core-hour that is \$0.100 spent against a \$0.049 estimate for the five, and \$0.084 against \$0.077 for the finer mesh. Money figures are derived from the hourly rate; they are not read from a metered bill.

law: blowing buys circulation, and circulation grows with the square root of jet momentum, so the system pays back fastest at low blowing rates. At $C_\mu = 0.40$ the jet provides 0.20 of lift coefficient directly and 1.06 indirectly, through the pressure field it induces on the aerofoil — five times the direct part, and that amplification is the point of the configuration.

Lead Engineer. Size the system at the knee, not at the top. Going from $C_\mu = 0$ to 0.05 delivers 0.42 of lift coefficient; quadrupling jet momentum from 0.10 to 0.40 delivers only another 0.65. Budget analysis time accordingly: this flow becomes markedly harder to solve as blowing increases, so a strongly blown case is not a mildly blown case with a different inlet number.

Lead Numericist. Confidence differs sharply between the two tables. Table 1 is arithmetic against a published expression, exact to the digits shown. Table 2 has not reached the target set before running: the turbulence equation imbalance grows monotonically with blowing, from 3.46×10^{-6} at the reference to 1.47×10^{-4} at the strongest jet, a factor of 43. The lift is still moving at the end of every run, by the amount in the movement column, which is why that column is quoted at all — a settling indicator is not a converged one. In every one of the five

conditions the solver also had to clip the turbulence energy back to zero wherever it went negative, on 81 to 98% of iterations and continuously through the final one, so these fields are held non-negative by that clipping rather than converged free of it; on the companion mesh it is 99.4%. The finer companion mesh has completed, and it does not close

Figure notes

The section. Two-dimensional wing section, 1.0 m chord, symmetric, at zero incidence in a 10.0 ms^{-1} stream, with a 5.0 mm slot at the trailing edge blowing a sheet of air 30° below the chord line. C_μ is the jet's momentum divided by the oncoming stream's dynamic pressure times the chord. Because the section is symmetric at zero incidence it makes no lift at all without blowing.

Lift against blowing. All five conditions share one grid of 39984 cells. The four blown ones differ from one another only in how hard the slot blows, so differences among those four are caused by the blowing and nothing else; the fifth is a reference with the slot closed, and is not the same wing with the jet turned down to zero. The first 0.05 of jet momentum buys a disproportionate share of the lift, which is the square-root law read off the chart.

the mesh question: it is a different grid topology on a different reference area, run as a diagnostic, so it is not a refinement of the five and no grid-refinement study has been carried out. The mesh contribution to the uncertainty on Table 2 therefore remains unquantified.

The flow picture. Red is air moving faster than the oncoming stream, blue slower, and white the same speed, with the dashed contour marking exactly the oncoming speed. The dark line traces the fast sheet leaving the slot; it bends the oncoming flow downward, and that bending is what lifts the wing, there being no hinged flap anywhere in the model. Contours and streamlines are drawn on a display grid for legibility, while every number quoted on this sheet is read from the solver's own output.

Where the sheet goes. The slot aims the sheet 30° below the chord line and the oncoming flow turns it, so it travels well above where a free jet of the same angle would go. That turning is the force: the jet pushes the air down and the air pushes the jet, and the wing, up. The trace stops where the sheet can no longer be told apart from the surrounding air and is not extrapolated beyond it.